Final Report: Multimodal Deep Learning Search Engine for E-Commerce Fashion

Course: DSCI471 - Deep Learning

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Abstract

We built and evaluated a multimodal fashion product search system that maps natural-language queries to product images using a dual-encoder architecture trained with contrastive learning. A TF-IDF keyword baseline indexes the same catalog using rich product text made from structured attributes plus JSON descriptions. On a held-out test gallery of 4,427 products, TF-IDF outperforms our final dual-encoder (v4) on Top-1 accuracy for all four query styles. With the largest gaps on templated queries (0.523 vs 0.174) and brand queries (0.651 vs 0.140). The dual-encoder remained competitive on shopper-style queries at Top-5 (0.241 vs 0.216) and MRR (0.167 vs 0.161), showing that cross-modal retrieval can place relevant products in the visible result set even when rank-1 accuracy is low.

The main result is not that deep learning "failed." Instead, the experiment shows a real tradeoff: keyword search is very strong when the index contains rich text while dual-encoders are the appropriate architecture when the retrieval target is an image gallery or when product text is incomplete. We documented the full data pipeline, exploratory analysis, v1-to-v5 model progression, final evaluation, limitations, and reproducibility steps.

Keywords: Multimodal retrieval, dual-encoder, contrastive learning, fashion e-commerce, TF-IDF baseline

1. Introduction

1.1 Motivation

E-commerce search often relies on exact keyword matching against product names, categories, and different metadata. However, shoppers frequently describe items in natural human language. For example, "a flowy floral dress with cap sleeves" or "men's navy casual shirt." If the search engine depends only on exact word overlap, retrieval quality drops when shoppers use different phrasing from the catalog.

However, multimodal deep learning offers a different approach. A dual-encoder model can learn a shared embedding space where text queries and product images are directly comparable. If the model learns useful visual-text alignment, it can retrieve images even when there is no exact keyword match at query time.

1.2 Research Question

"Can a dual-encoder deep learning model, trained on paired fashion images and text descriptions, outperform traditional keyword-based e-commerce search?**"**

1.3 Contributions

1. A full preprocessing pipeline for 44,265 usable fashion products with images, structured metadata, generated captions, and text indices.
2. Exploratory analysis of product categories, article types, colors, image samples, and product-text length.
3. A dual-encoder retrieval model using EfficientNetB0 for images and MiniLM sentence embeddings for text.
4. A strong TF-IDF baseline and shared evaluation framework reporting Top-1, Top-5, MRR, and Precision@5.
5. An ablation path from v1 to v5 showing why the final v4 architecture was selected.
6. A critical interpretation explaining why TF-IDF wins Top-1 and where the dual-encoder is still useful.

1.4 Related Work and Definitions

Dual-encoder retrieval. CLIP popularized contrastive pretraining of image and text encoders on large image-text datasets. The core idea was to map paired images and captions that are close together in an embedding space while still pushing mismatched pairs apart. Our project applies that same retrieval pattern, however on a course-project scale of fashion catalog.

Fashion and e-commerce search. Fashion retrieval systems often use product attributes, image features, or multimodal embeddings for search and recommendations. So with this in mind, the Kaggle Fashion Product Image dataset is well suited for this because each item has both product photo and metadata such as color, article type, gender, season, and usage.

Keyword baselines. TF-IDF and BM25 remain as strong baselines for catalog search when product text is available for users. Our TF-IDF baseline is intentionally strong because it indexes on product_text, which combines generated captions and JSON descriptions together. This makes the comparison honest that the deep model is tested against a practical classical system instead of a weak strawman.

2. Data Collection and Exploratory Analytics

2.1 Source Dataset

We used the Fashion Product Image dataset (Aggarwal, Kaggle), which contains approximately 44,400 catalog products. Each product has:

- A product image stored as images/{id}.jpg
- Structured fields from styles.csv
- Optional JSON product information from styles/{id}.json
- Attributes including gender, master category, subcategory, article type, base color, season, usage, and product display name

2.2 Preprocessing Pipeline

The preprocessing pipeline is implemented in src/prepare_data.py

1. Load product metadata from styles.csv
2. Keep only rows with matching image files.
3. Load JSON descriptions when present.
4. Generate templated product captions using src/captions.py
5. Build product_text by combining generated captions and JSON descriptions.
6. Drop rare articleType values with fewer than 10 products.
7. Create stratified train, validation, and test splits by articleType.

2.3 Training Split

Final Corpus: 44,265 usable products

Split	**Products**	**Purpose**
Train	35,412	Dual-encoder training
Validation	4,426	Development and ablation selection
Test	4,427	Final held-out evaluation gallery

Split ratio is 80/10/10 with random seed 42. Outputs are saved to data/processed/train.csv, val.csv, test.csv, product.csv, and pairs.csv

2.4 Dataset Statistics

The final processed dataset contains:

Statistic	**Value**
Products after filtering	44,265
Master categories	6
Article types	107
Base colors	46
Products with loaded JSON descriptions	43,983
Median product_text length	674 characters
Mean product_text length	638 characters

The catalog is imbalance, which matters for retrieval. The largest article types are T-shirt (7,066), Shirts (3,215), Casual Shoes (2,845), Watches (2,542), Sport Shoes (2,036), Kurtas (1,844), Tops (1,762), Handbags (1,759), Heels (1,323), and Sunglasses (1,073). Apparel dominates the dataset with 21,327 products, followed by Accessories (11,243) and Footwear (9,219).

Figure 1 - Article-type distribution after preprocessing:

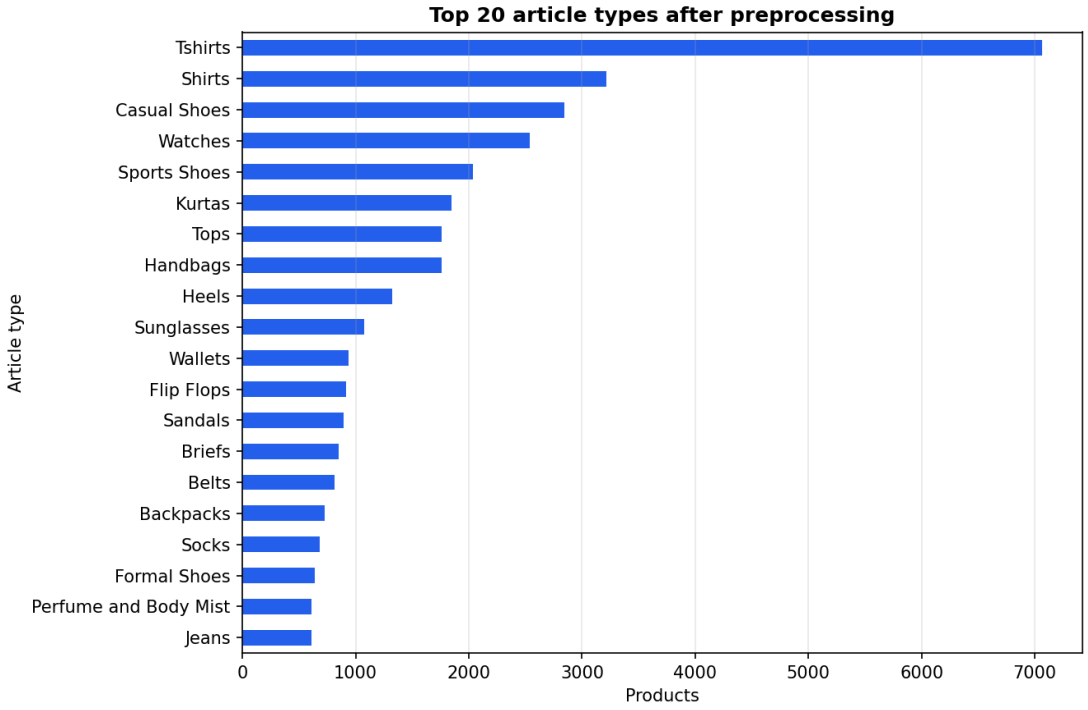


Figure 2 - Master category distribution:

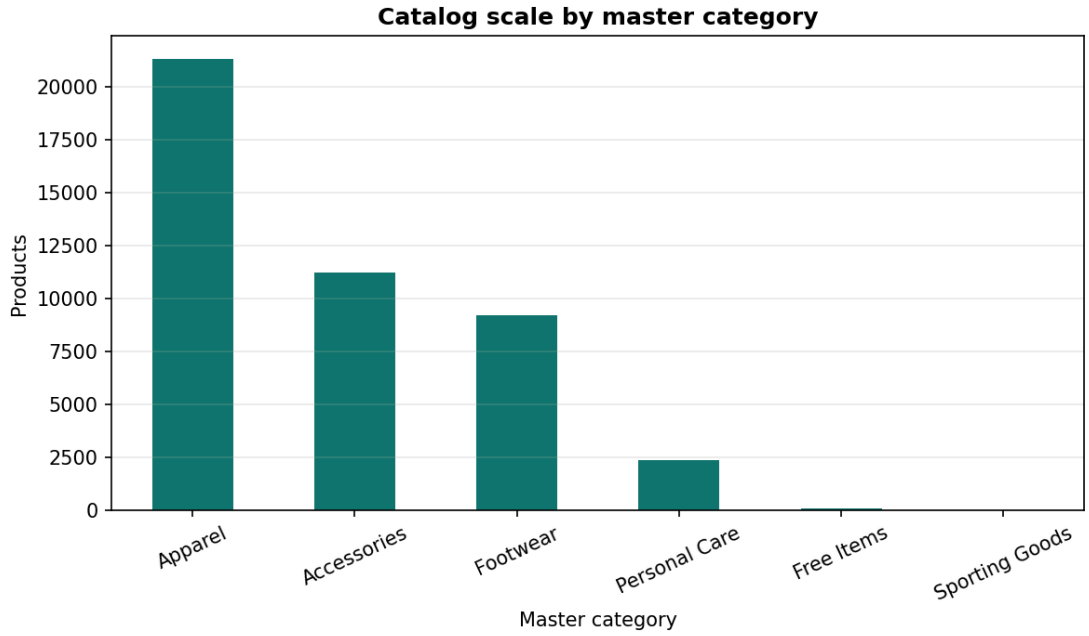
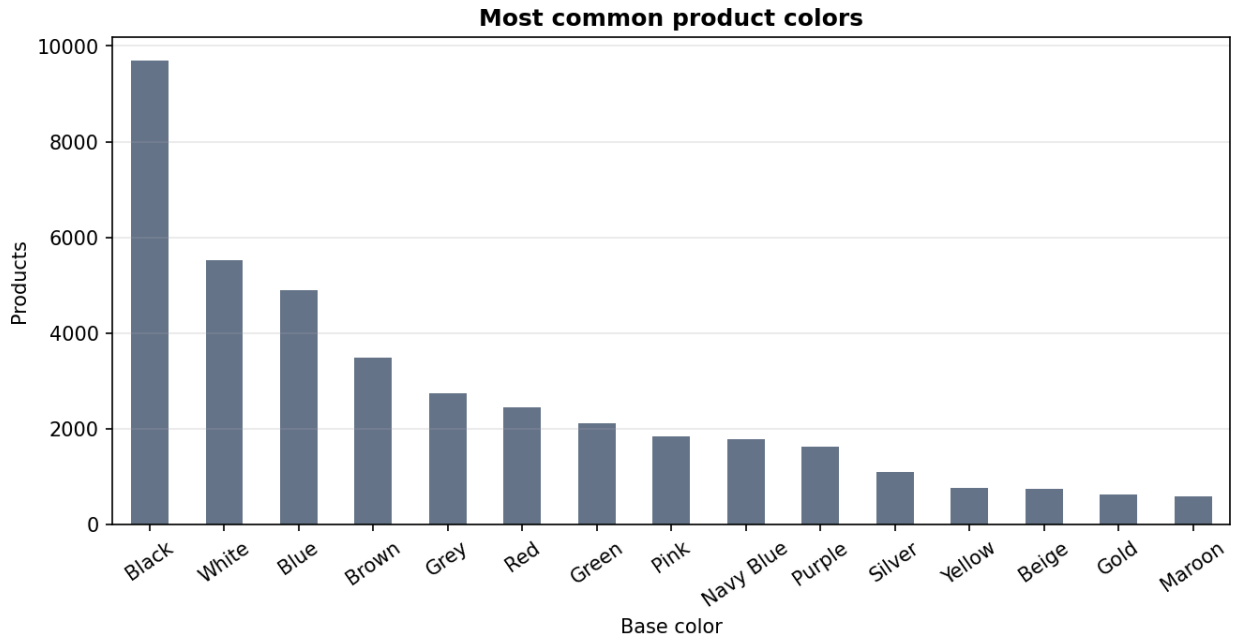


Figure 3 - Color distribution:

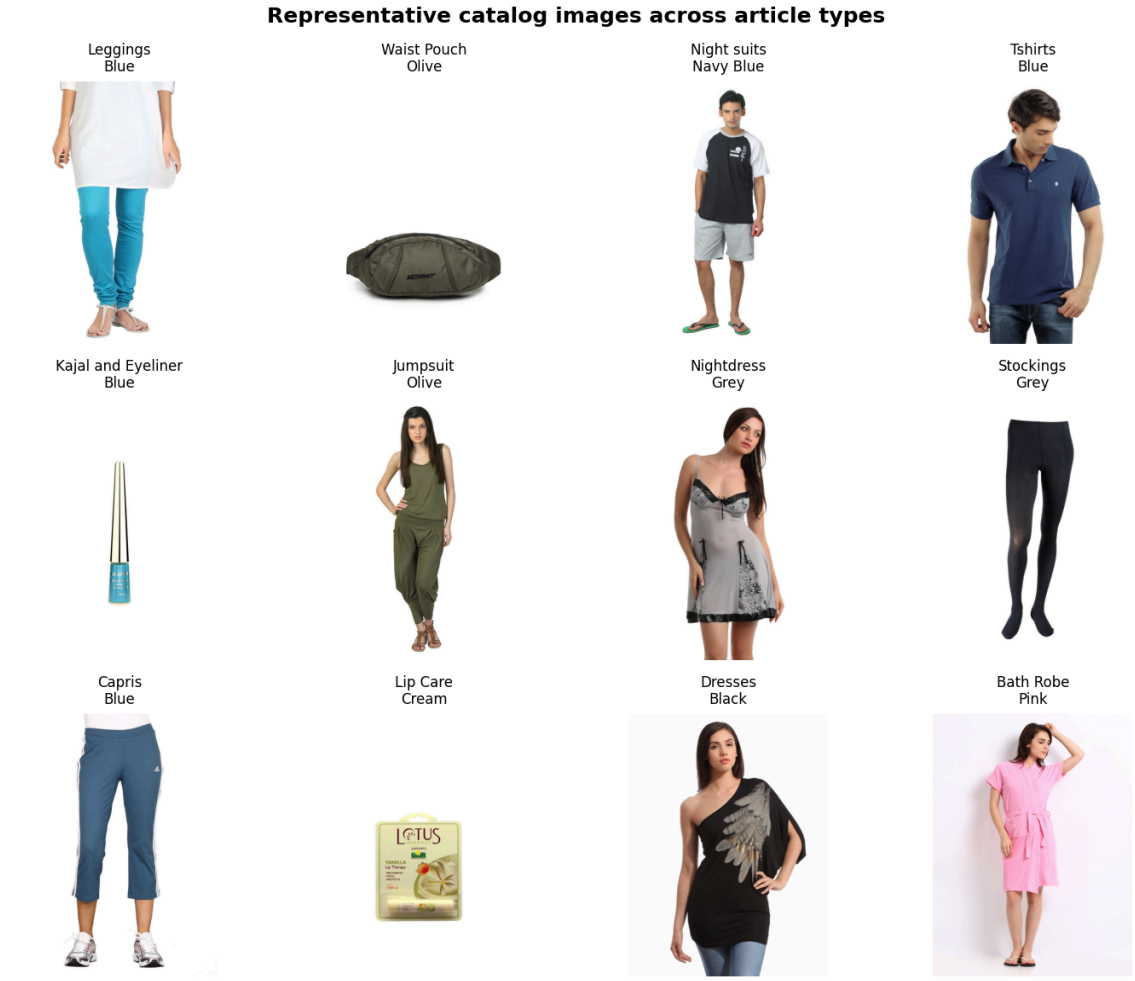


Black, white, and blue products are especially common. This can create ambiguity for short queries such as “black shoes” or “blue shirt” because many products can be visually and semantically plausible.

2.4 Image and Text Characteristics

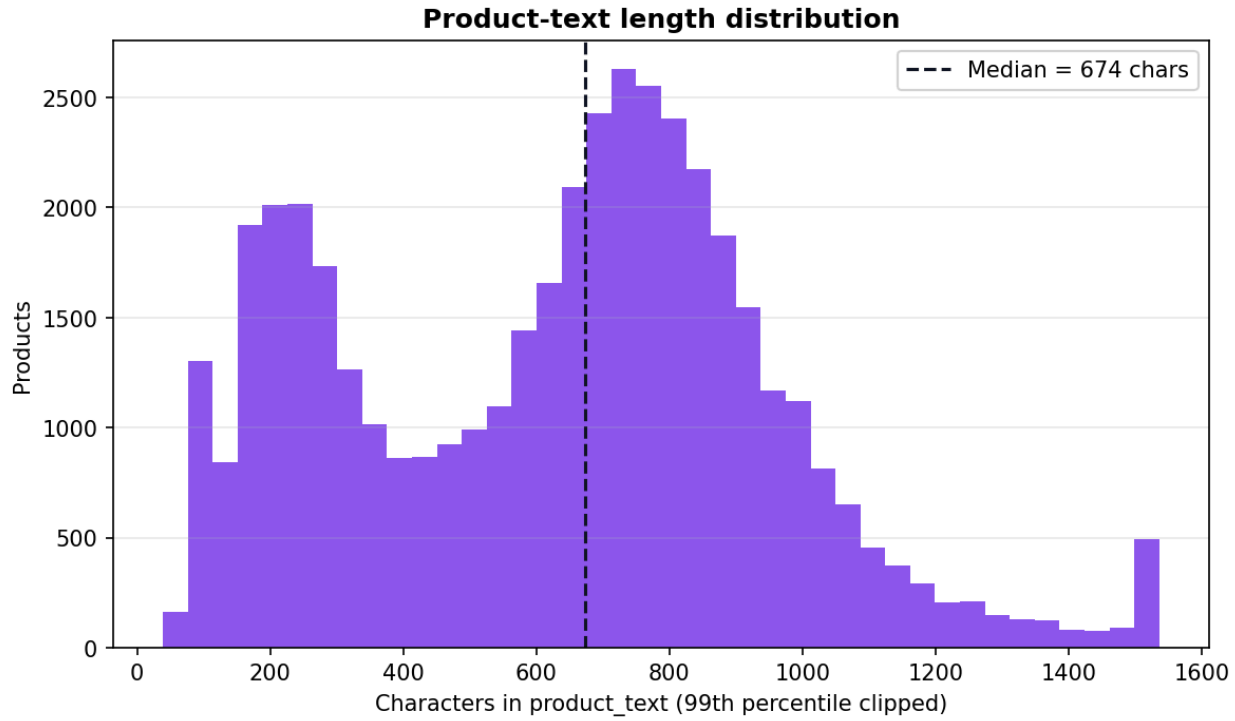
The product images are clean catalog photos, usually centered on a single product with a plain background. This supports CNN feature extraction because the relevant object is usually visible and not heavily occluded. Still, the images often lack details that appear in product text, such as brand names, intended season, or product-specific titles.

Figure 4 - Representative product images:



The text side is also uneven, as some rows have long JSON descriptions, while others mainly contain generated attribute captions. We therefore use a robust baseline (product_text) for TF-IDF and use generated captions for multimodal pairing.

Figure 5 - Product-text length distribution:



2.5 Evaluation Query Styles

Each test product generates one query per style:

Style	Description	Example
templated	Full generated attribute caption	A men's navy blue shirt, for casual wear in fall. Turtle Check Men Navy Blue Shirt.
shopper	Short natural phrase	men’s navy blue shirt
brand	Product display name	Turtle Check Men Navy Blue Shirt
short	Color plus article type	Navy shirt

The same query generator is used for both TF-IDF and the dual-encoder, so the final comparison uses identical test cases.

3. Methodology and Model Justification

3.1 Final Dual-Encoder Architecture (v4)

Component	Specification
Image encoder	EfficientNetB0, ImageNet pretrained
Image input size	224 x 224 x 3
Projection head	Dense(512, GELU) -> Dropout(0.1) -> Dense(384)
Text encoder	Frozen sentence-transformers/all-MiniLM-L6-v2
Embedding dimension	384
Normalization	L2-normalized image and text embeddings
Similarity	Dot product / cosine similarity
Loss	Symmetric contrastive loss, InfoNCE-style
Temperature	0.07

Implementation lives in src/model.py, with hyperparameters in src/config.py.

3.2 Why EfficientNetB0?

EfficientNetB0 was selected because it gives a strong accuracy-to-compute tradeoff for a course-scale project. A larger CNN could improve visual representations, but would make training and reproduction harder on CPU laptops. EfficientNetB0 also comes with ImageNet-pretrained weights, giving the model general visual features such as edges, textures, shapes, and object parts before fine-tuning on fashion products.

The 224 x 224 input size matches the standard EfficientNetB0 configuration and keeps the image pipeline simple. Product photos in this dataset are mostly centered, so resizing is a reasonable spatial preprocessing step. We do not use sequence padding or temporal masking because the input is image-text pairs, not time series; the relevant spatial constraint is consistent resizing into the CNN input grid.

3.3 Why MiniLM Text Embeddings?

Early versions used a smaller scratch texting encoder, but v4 switched to all-MiniLM-L6-v2. This was the largest improvement in validation ablations: templated Recall@1 rose from 0.106 in v1 to 0.208 in v4. MiniLM gives compact 384-dimensional sentence embeddings, which match the target embedding size

and reduce computation. We froze MiniLM to avoid expensive transformer fine-tuning and to reduce overfitting risk given the synthetic nature of the captions.

3.4 Projection Head, Activation, Dropout, and Normalization

The image tower does not directly compare EfficientNet features to text embeddings. Instead, it learns a projection head:

- Dense(512) gives the model capacity to adapt ImageNet features to fashion attributes.
- GELU is a smooth nonlinearity commonly used in modern neural networks and works very well with transformer-style embedding spaces.
- Dropout(0.1) adds light regularization without overwhelming a small project head.
- Dense(384) maps image features into the same dimensionality as MiniLM text embeddings.
- L2 normalization makes dot products equivalent to cosine similarity, which is standard for retrieval embeddings.

The contrastive temperature of 0.07 sharpens the similarity distribution. Lower temperatures make the model focus more strongly on the correct image-text pair inside each batch. This is CLIP-like behavior and is appropriate when each batch contains many in-batch negatives.

3.5 Training Procedure

Training is implemented in src/train.py.

1. Phase 1: Freeze EfficientNetB0 and train only the project head for 4 epochs with Adam at a learning rate of 1e-3.
2. Phase 2: Unfreeze the last 20 EfficientNet layers and fine-tune for only 3 epochs with a learning rate of 1e-5.
3. Cache MiniLM text embeddings for training/validation captions under models/embeddings/.

This two-staged approach was intentionally designed for development. This is because the first stage learns a stable image-to-text projection without disrupting pretrained visual features. The second stage lightly adapts high-level EfficientNet layers to the fashion domain while avoiding large updates to the full CNN.

3.6 TF-IDF Baseline

The baseline uses TfidVectorizer(stop_words=”english”, max_features=50000) on the test-gallery product_text. Queries are then transformed into the same sparse vector space and ranked by cosine similarity.

This is strong but a realistic baseline. It performs text-to-text retrieval with access to captions and JSON descriptions. The dual-encoder performs text-to-image retrieval, which is harder because of brand names and descriptions that are not directly visible to the image.

3.7 Evaluation Protocol

For final evaluation:

- Gallery: all 4,427 held-out test products
- Queries: one query per product per query style
- Target: the exact product ID that generated the query
- Metrics: Top-1, Top-5, Mean Reciprocal Rank (MRR), Precision@5
- Code: src/evaluate.py and src/metrics.py

Development ablations use the validation split. Final results use only the test split.

4. Experiments and Model Selection

Richardson led the iterative ablation notebooks in notebooks/richardson_experiment/. Samii unified the final training, evaluation, and documentation in pipeline src/ and docs/.

Version	**Main Change**	**Outcome**
v1	Scratch text encoder, templated caption, full dataset	Established the first dual-encoder baseline
v2	Caption expansion with four query styles	More realistic query diversity, but lower templated recall
v3	Random query-style rotation	Did not clearly beat v2
v4	Frozen MiniML text encoder plus EfficientNet image tower	Best validation Recall@K; selected final model
v5	V4 plus caption rotation	Did not beat v4 on validation

4.1 Validation Ablation

For final evaluation:

Model	**R@1**	**R@5**	**R@10**
v1 - scratch test	0.106	0.332	0.488
v2 - caption expansion	0.080	0.271	0.425
v3 - rotation	0.068	0.247	0.387
v4 - MiniLM text	**0.208**	**0.523**	**0.668**

Pretrained text encoding was the most important improvement. Caption rotation did not automatically help because the generated query styles were still synthetic and sometimes too ambiguous.

Figure 6 - Validation Recall@1 progression:

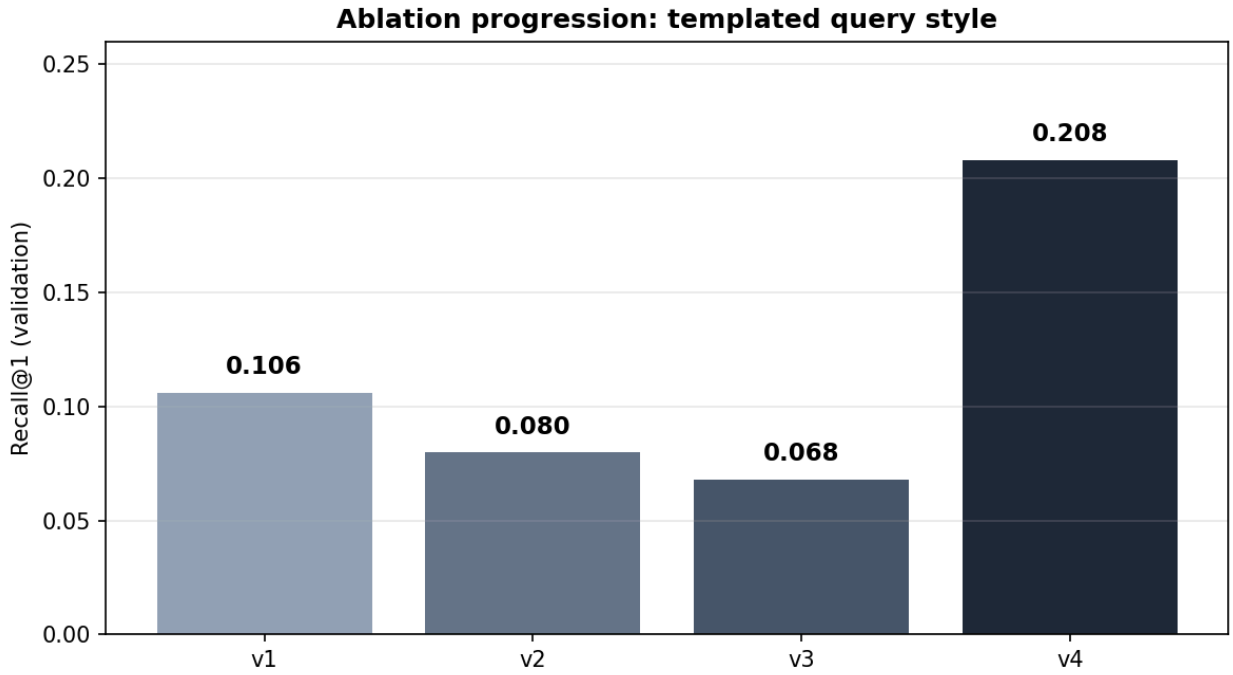


Figure 7 - v4 training loss:

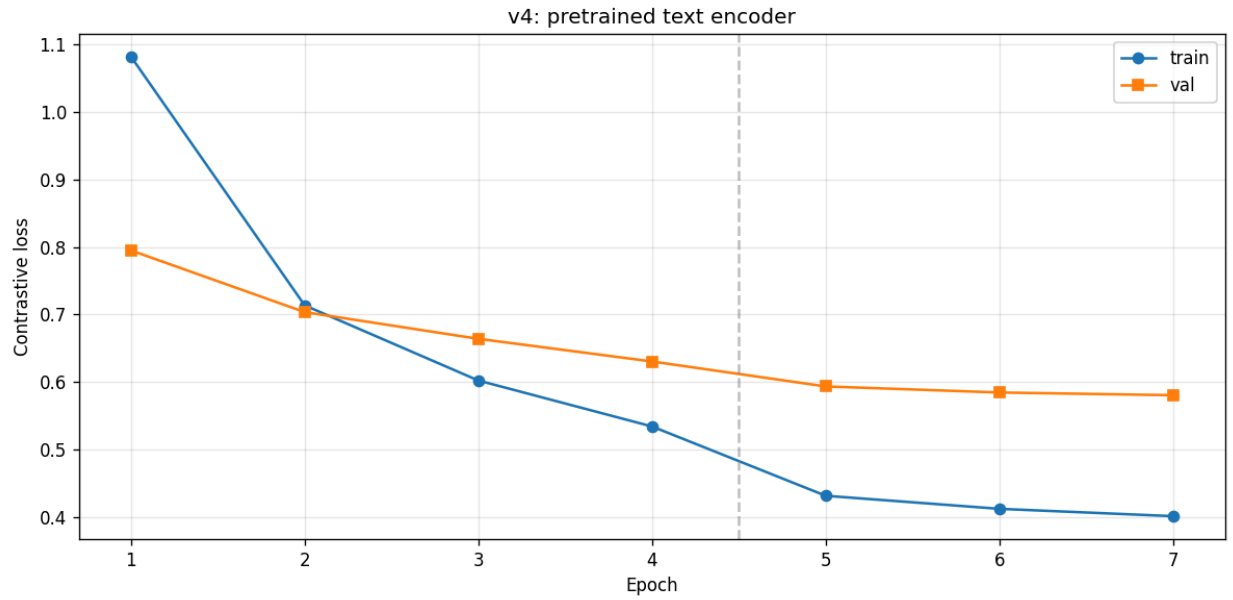


Figure 8 - v4/v5 comparison:

5. Results

5.1 Test-Set Metrics

Full metrics are stored in docs/reports/evaluation_results.csv

Model	Query Type	Top-1	Top-5	MRR	Precision@5
TF-IDF	templated	0.523	0.805	0.649	0.161
TF-IDF	shopper	0.084	0.216	0.161	0.043
TF-IDF	brand	0.651	0.891	0.756	0.178
TF-IDF	short	0.073	0.201	0.146	0.040
Dual-encoder (v4)	templated	0.174	0.462	0.314	0.092
Dual-encoder (v4)	shopper	0.074	0.241	0.167	0.048
Dual-encoder (v4)	brand	0.140	0.402	0.270	0.080
Dual-encoder (v4)	short	0.052	0.176	0.126	0.035

Figure 9 - Test-set retrieval metrics:

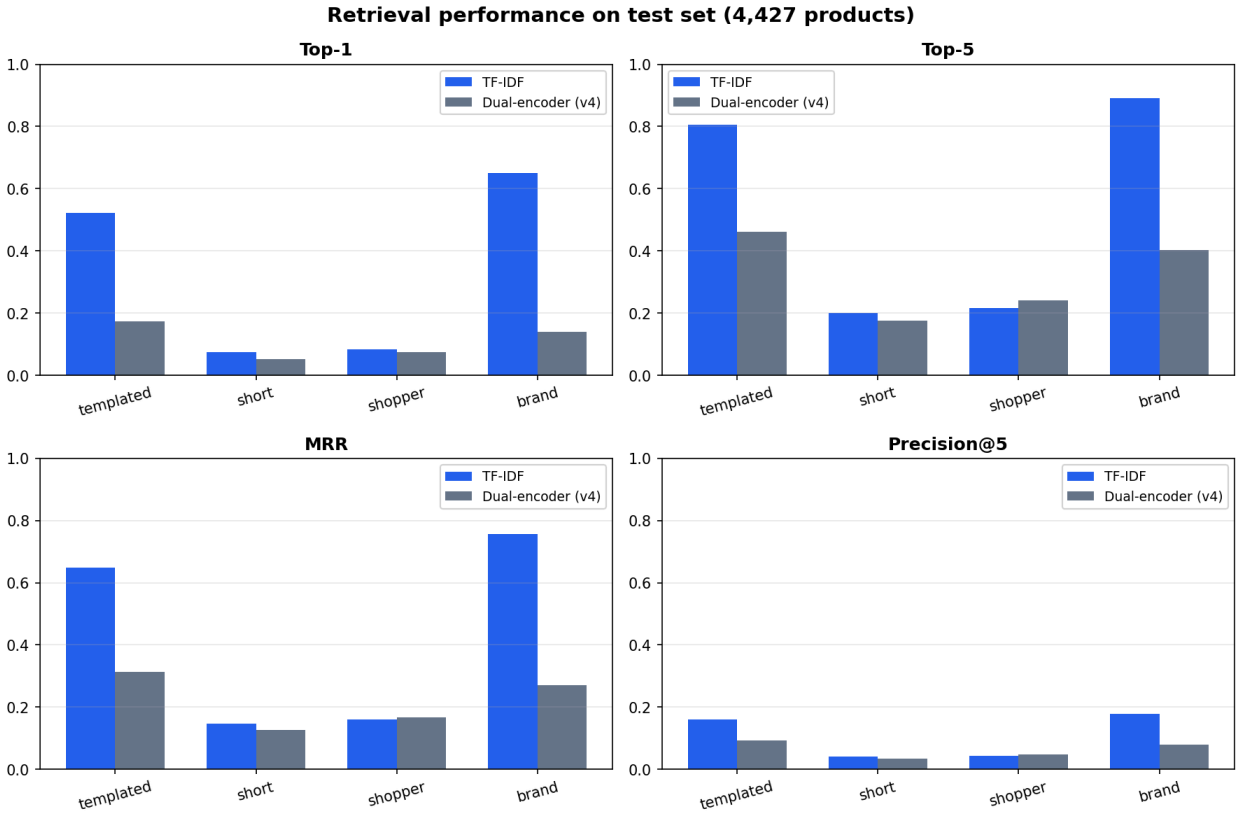
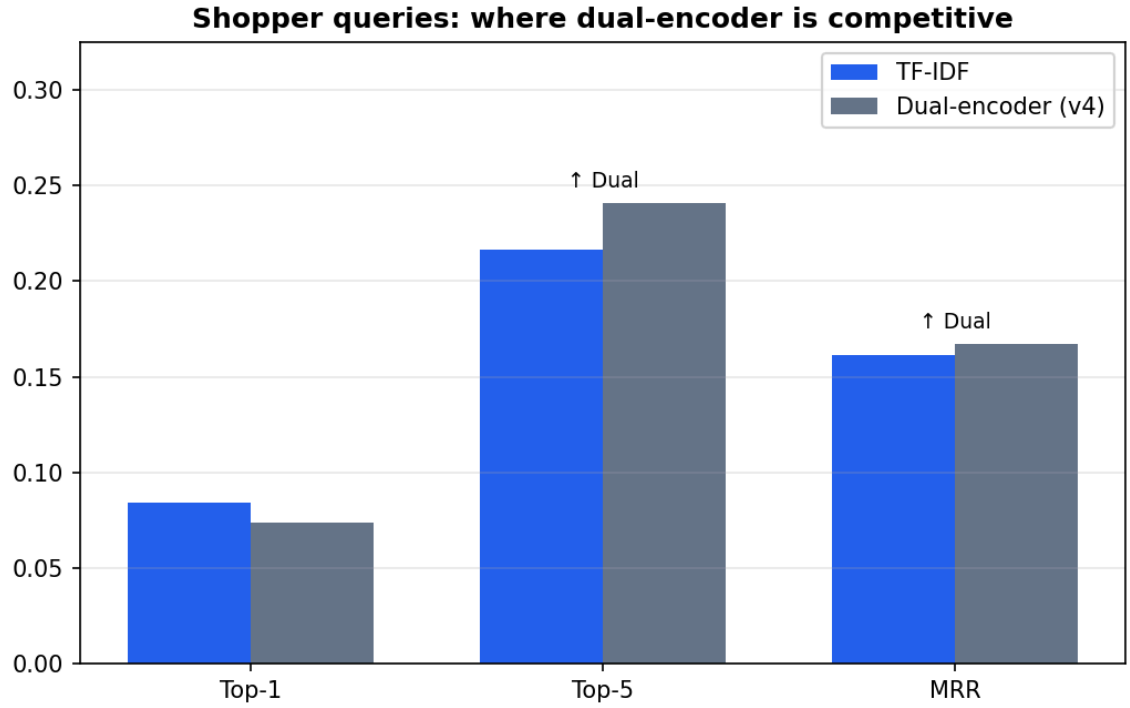


Figure 10 - Shopper-quer comparison:



5.2 Answer to the Research Question

The answer is **NO** for Top-1 retrieval in this test setup. The dual-encoder does not outperform TF-IDF on rank-1 accuracy for any query style.

Query Type	**TF-IDF Top-1**	**Dual Top-1**	**Difference**
templated	0.523	0.174	-0.349
shopper	0.084	0.074	-0.010
brand	0.651	0.140	-0.511
short	0.073	0.052	-0.022

The result is strongest for brand and templated queries because those queries closely match the text indexed by TF-IDF. Brand names and product titles were often not visible in the image, so the dual-encoder cannot recover that information from pixels alone.

The dual-encoder's best relative result is on the shopper queries, where it beats TF-IDF on Top-5 and MRR. That means it sometimes places the exact item higher in the shortlist even when it misses the top position.

5.3 Qualitative Retrieval Examples

Figure 10 - Dual-encoder success on a templated query:

Figure 11 - Brand-query head-to-head:

Figure 12 - Shopper-query failure case:

These examples show the pattern behind the metrics. The model can align broad visual categories, colors, and product types, but exact product identity is difficult when many items share the same visual attributes with one another.

6. Interpretation and Critical Thinking

6.1 Why TF-IDF Wins

TF-IDF wins because the evaluation contains queries that are often text-identical or text-adjacent to the product metadata. Brand queries are product display names. Templated queries are generated from the same structured fields that appear in product_text. Because TF-IDF searches text against text, it receives direct evidence that the dual-encoder does not have at image-index time.

However, this does not invalidate the dual-encoder. It clarifies the deployment settings. If the product text is available and has a high quality image, a text retrieval system is extremely competitive with it. If the searchable index is only images or weakly described, the dual-encoder is more efficient and appropriate.

6.2 Overfitting, Underfitting, or Architecture Ceiling?

The final appears to hit a data/evaluation ceiling more than a simple overfitting pattern. Several observations were able to support this:

- v4 improves substantially over v1 on validation Recall@1, so the architecture learned useful alignment.
- TF-IDF's biggest wins occur where text identity matters the most: brand and templated queries.
- Shopper and short queries are ambiguous because many products share the same color and article type.
- The single-positive evaluation counts only the exact product as correct, even when visually similar products may be reasonable search results.

There may still be some underfitting in the image-text alignment because the model was trained with limited compute, batch size of 64, and a frozen text tower. However, simply training longer would not solve brand-name retrieval from images. The deeper limitation is that the image embedding cannot encode metadata that is not visually observable.

6.3 What Ablations Teaches

The v1 to v5 experiments showed that the text representation quality mattered the most compared to caption rotation. MiniLM improved semantic structure into the text space. In addition, random query-style rotation did not reliably improve retrieval because the generated queries remained synthetic and sometimes underspecified. This suggests that future gains need to come from better model supervision, hard negatives, real queries from users, and domain-tuned multimodal pretraining instead of more epochs.

6.4 When Deep Learning is Still Helpful

The dual-recorder still remains useful for:

- Image-only galleries where TF-IDF cannot be fully applied.
- Visual similarities searches and cross-modal matching.
- Shortlist retrieval where Top-5 matters more than Top-1 ranking.
- Hybrid search system where TF-IDF retrieves candidates and a visual model reranks them.

6.5 Project Limitations

There are limitations to our project. Such as:

1. **Synthetic queries:** Queries that are generated from catalog fields, not actual real user logs
2. **Single positive per query:** The evaluation only counts the exact product as correct, even though visually similar products may be relevant instead.

3. **Ambiguous fashion attributes:** Queries like “black shoes”, etc. can match to many different products.
4. **Frozen text encoder:** MiniLM was not fine-tuned on fashion-specific search behavior.
5. **Limited compute:** Training uses a practical CPU-friendly setup instead of large-batch contrastive learning.
6. **Metadata not being visible:** Brand names, titles, and descriptions are not available in most of the pixels.

7. Conclusion and Potential Future Work

As a group we implemented a complete multimodal fashion retrieval project. Done with data preparation, exploratory analysis, caption generation, EfficientNet/MiniLM dual-encoder training, TF-IDF baseline retrieval, ablation tracking, and final evaluation on a test set.

The final conclusion is very nuanced. TF-IDF is the stronger model for rank-1 search systems when rich product text is available for development. The dual-encoder is still a valid cross-modal retrieval modal and performs competitively well on shopper-query Top-5 and MRR. The project therefore answers the research question while also still explaining the conditions under which each search method is appropriate.

We don’t plan on continuing development after the class ends. However potential future work could include:

- Fine-tuning the text encoder on fashion captions and queries
- Add hard negative mining for visually similar products
- Increase effective contrastive batch size
- Use CLIP-style fashion-domain pretraining
- Evaluate with real user search logs and click-through data
- Build a hybrid system: TF-IDF shortlist followed by visual reranking